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4

EQUILIBRIUM GAS PROPERTIES

4.1 Spatial properties

The results that were obtained in §3.4 for the equilibrium distribution function f_0 may be summarized as follows:

$$f_0 = (\beta^3/\pi^{3/2})\exp(-\beta^2 c'^2),$$

where (4.1)

$$\beta = (2RT)^{-1/2} = \{m/(2kT)\}^{1/2}.$$

The fraction of the molecules that are located within a velocity space element of volume $\mathrm{d}\boldsymbol{c}$ and located at $\boldsymbol{c}'$ follows from eqn (3.1) as

$$\frac{\mathrm{d}N}{N} = \frac{\mathrm{d}n}{n} = (\beta^3/\pi^{3/2})\exp(-\beta^2 c'^2)\,\mathrm{d}\boldsymbol{c}. \tag{4.2}$$

The peculiar or thermal velocity $\boldsymbol{c}'$ is equal to $\boldsymbol{c} - \boldsymbol{c}_0$ so that, in Cartesian coordinates (u, v, w), the fraction of molecules with velocity components from u to $u + \mathrm{d}u$, v to $v + \mathrm{d}v$, and w to $w + \mathrm{d}w$ is

$$\frac{\mathrm{d}n}{n} = (\beta^3/\pi^{3/2})\exp[-\beta^2\{(u-u_0)^2 + (v-v_0)^2 + (w-w_0)^2\}]\,\mathrm{d}u\,\mathrm{d}v\,\mathrm{d}w. \tag{4.3}$$

For *polar coordinates* (c', θ, ϕ) in a frame of reference moving with the stream velocity, the volume of the velocity space element is

$$c'^2 \sin\theta\,\mathrm{d}\theta\,\mathrm{d}\phi\,\mathrm{d}c'.$$

The fraction of molecules with speed between c' and $c' + \mathrm{d}c'$, which make an angle between θ and $\theta + \mathrm{d}\theta$ with the polar direction, and have an azimuth angle between ϕ and $\phi + \mathrm{d}\phi$ is, therefore,

$$\frac{\mathrm{d}n}{n} = (\beta^3/\pi^{3/2})c'^2\exp(-\beta^2 c'^2)\sin\theta\,\mathrm{d}\theta\mathrm{d}\phi\mathrm{d}c'. \tag{4.4}$$

The fraction of molecules with speeds between c' and $c' + \mathrm{d}c'$, irrespective of direction, is obtained from eqn (4.4) by integrating ϕ between the limits 0 to 2π and θ from 0 to π, to give

$$(4/\pi^{1/2})\beta^3 c'^2\exp(-\beta^2 c'^2)\,\mathrm{d}c'. \tag{4.5}$$

A distribution function $f_{c'}$ may be defined such that the fraction of molecules with speeds between c' and $c' + \mathrm{d}c'$ is $f_{c'}\mathrm{d}c'$. Eqn (4.5) shows that

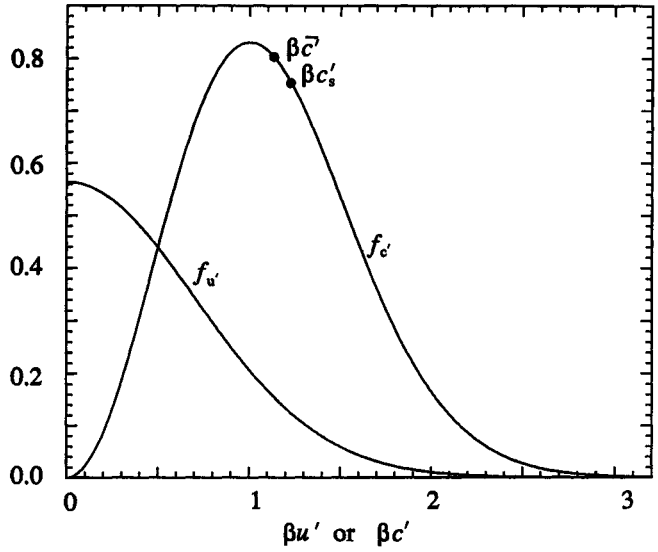


Fig. 4.1 Equilibrium distribution functions for the molecular speed and for a molecular velocity component.

$$f_{c'} = (4/\pi^{1/2})\,\beta^3 c'^2 \exp(-\beta^2 c'^2). \tag{4.6}$$

The function $f_{c'}$ is plotted in Fig. 4.1. It is zero when c' is zero, increases to a maximum value when $\beta\, c'$ is unity, and then decreases as c' increases. The parameter β is, therefore, the reciprocal of the *most probable molecular thermal speed* c'_m, i.e.

$$c'_m = 1/\beta. \tag{4.7}$$

The average of any quantity that depends on the molecular speed may be obtained through the application of eqn (3.3) to the thermal speed distribution function $f_{c'}$. The *average thermal speed* $\overline{c'}$ is

$$\overline{c'} = \int_o^\infty c' f_{c'} dc' = (4/\pi^{1/2})\beta^3 \int_0^\infty c'^3 \exp(-\beta^2 c')\, dc',$$

or, referring to the standard integrals in Appendix B,

$$\overline{c'} = 2/(\pi^{1/2}\beta) = (2/\pi^{1/2})\, c'_m. \tag{4.8}$$

The mean square thermal speed has already been determined in eqn (3.48) through the evaluation of β in terms of T. The root mean thermal speed c'_s follows as

$$c'_s = (3/2)^{1/2}(1/\beta) = (3/2)^{1/2} c'_m = (3\,\pi/8)^{1/2}\,\overline{c'}. \tag{4.9}$$

The ordering $c'_s > \overline{c'} > c'_m$ is a consequence of the high-speed tail of the distribution function. The fraction of molecules with thermal speed above some value c' is given by the integral of eqn (4.5) from c' to ∞, i.e.

$$(4/\pi^{1/2})\beta^3 \int_{c'}^{\infty} c'^2 \exp(-\beta^2 c'^2)\, dc'.$$

Again using the standard integrals of Appendix B, this fraction becomes

$$1 + (2/\pi^{1/2})\beta c' \exp(-\beta^2 c'^2) - \mathrm{erf}(\beta c'). \tag{4.10}$$

The *error function*

$$\mathrm{erf}(a) = \frac{2}{\pi^{1/2}} \int_0^a \exp(-x^2)\, dx \tag{4.11}$$

will be encountered frequently. Appendix B presents tabulated values of this function, together with its limiting values, series representations, and a rational approximation.

The fraction of molecules with a velocity component within a given range, irrespective of the magnitude of the other components, is obtained by integrating eqn (4.2) over these other components. For example, the fraction of molecules with a thermal velocity component in the x-direction of between u' and $u' + du'$ is

$$(\beta/\pi^{1/2})^3 \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \exp\{-\beta^2(u'^2 + v'^2 + w'^2)\}\, du'\, dv'\, dw',$$

or

$$(\beta/\pi^{1/2}) \exp(-\beta^2 u'^2)\, du'. \tag{4.12}$$

The distribution function for a thermal velocity component is, therefore,

$$f_{u'} = (\beta/\pi^{1/2}) \exp(-\beta^2 u'^2). \tag{4.13}$$

This function is also plotted in Fig. 4.1. Since the velocity distribution function f is spherically symmetric about the point representing the stream velocity, the most probable value of a particular thermal velocity component is zero. The average of the thermal velocity components in the x-direction averaged over only those molecules moving in the positive x-direction is

$$\begin{aligned} \int_0^{\infty} u' f_{u'}\, du' \Big/ \int_0^{\infty} f_{u'}\, du' &= 2(\beta/\pi^{1/2}) \int_0^{\infty} u' \exp(-\beta^2 u'^2)\, du' \\ &= 1/(\pi^{1/2}\beta) \end{aligned} \tag{4.14}$$

or, comparing this result with eqn (4.8), $\overline{c'}/2$.

Cylindrical coordinates in a frame of reference moving with the stream may be defined with a thermal velocity component c'_x in the axial direction, a component c'_n in the radial direction and an azimuth angle ϕ. The volume of the velocity space element is

$$c'_n \, d\phi \, dc'_n \, dc'_x$$

and the fraction of molecules with an azimuth angle between ϕ and $\phi+d\phi$, a radial velocity component between c'_n and $c'_n+dc'_n$, and an axial component between c'_x and $c'_x+dc'_x$ is

$$\frac{dn}{n} = (\beta^3/\pi^{3/2}) c'_n \exp\{-\beta^2(c'^2_x + c'^2_n)\} \, d\phi \, dc'_n \, dc'_x. \tag{4.15}$$

Therefore, the distribution function for the axial component of the thermal velocity is the same as that for the velocity component in any Cartesian direction, while the distribution for the radial component is

$$f_{c'_n} = 2\beta^2 c'_n \exp(-\beta^2 c'^2_n). \tag{4.16}$$

It is readily shown that the most probable radial velocity component is $1/(2^{1/2}\beta)$, while the average radial component is $\pi^{1/2}/(2\beta)$.

The high-speed 'tail' of the distribution functions is relevant to many practical problems. The fraction of molecules with thermal speeds greater than c_c is, from eqn (4.6),

$$\frac{dn}{n} = \frac{4\beta^3}{\pi^{1/2}} \int_{c_c}^{\infty} c'^2 \exp(-\beta^2 c'^2) \, dc',$$

or

$$\frac{dn}{n} = 1 - \mathrm{erf}(\beta c_c) + \frac{2}{\pi^{1/2}} \beta c_c \exp(-\beta^2 c_c^2). \tag{4.17}$$

For large values of βc_c, the exponential term is dominant and, from eqn (B34),

$$\frac{dn}{n} = \frac{1}{\pi^{1/2}} \exp(-\beta^2 c_c^2) \left(2\beta c_c + \frac{1}{\beta c_c} - \frac{1}{2\beta^3 c_c^3} + \ldots \right). \tag{4.18}$$

The fraction is approximately 0.046 for $\beta c_c=2$ and 0.00044 for $\beta c_c=3$, but declines to 8.0×10^{-11} at $\beta c_c=5$ and 4.2×10^{-43} at $\beta c_c=10$.

For a gas mixture in equilibrium, the equations of this section may be applied separately to each of the molecular species in the mixture.

4.2 Fluxal properties

We will now consider the flux of molecular quantities across a surface element in an equilibrium gas. The stream velocity $\boldsymbol{c}_0$ is inclined at the angle θ to the unit normal vector $\boldsymbol{e}$ to the surface element, as shown in Fig. 4.2. Without loss of generality, we may choose Cartesian coordinates such that the stream velocity lies in the xy-plane and the surface element lies in the yz-plane, with the x-axis in the negative $\boldsymbol{e}$ direction.

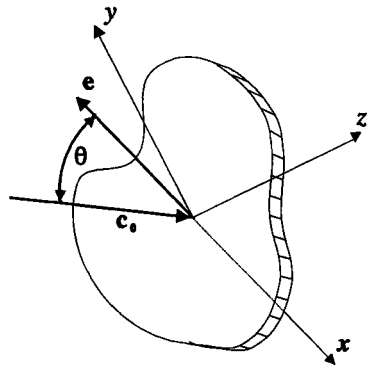


Fig. 4.2 Coordinate system for the analysis of molecular flux across a surface element.

Each molecule has velocity components

$$u = u' + c_0 \cos\theta,$$
$$v = v' + c_0 \sin\theta, \tag{4.19}$$

and

$$w = w'.$$

Therefore, from eqn (1.20), the *inward* (i.e. in the negative $\boldsymbol{e}$ or positive x-direction) flux of some quantity Q is

$$n\overline{Qu},$$

or

$$n \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{0}^{\infty} Quf \,du\,dv\,dw, \tag{4.20}$$

where consideration has been limited to those molecules moving in the positive x-direction. For an equilibrium gas, the function f_0 may be substituted from eqn (4.1) to give the inward flux of the quantity Q across the element as

$$\frac{n\beta^3}{\pi^{3/2}} \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{0}^{\infty} Qu \exp\{-\beta^2(u'^2 + v'^2 + w'^2)\}\,du\,dv\,dw$$

per unit area per unit time. Eqn (4.19) enables this result to be written in terms of the stream velocity and the thermal velocity components only, i.e.

$$\frac{n\beta^3}{\pi^{3/2}} \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-c_0\cos\theta}^{\infty} Q(u' + c_0\cos\theta) \exp\{-\beta^2(u'^2 + v'^2 + w'^2)\}\,du'\,dv'\,dw' \tag{4.21}$$

The *inward number flux* $\dot{N}_i$ to the element is obtained by setting $Q = 1$ in eqn (4.21). The variables in the integral may be separated to give

$$\dot{N}_i = \frac{n\beta^3}{\pi^{3/2}} \int_{-\infty}^{\infty} \exp(-\beta^2 w'^2)\,dw' \int_{-\infty}^{\infty} \exp(-\beta^2 v'^2)\,dv' \times \int_{-c_0\cos\theta}^{\infty} (u' + c_0\cos\theta)\exp(-\beta^2 u'^2)\,du'.$$

The standard integrals of Appendix B enable this to be written

$$\beta\dot{N}_i/n = [\exp(-s^2\cos^2\theta) + \pi^{1/2} s\cos\theta\{1 + \operatorname{erf}(s\cos\theta)\}]/(2\pi^{1/2}), \tag{4.22}$$

where

$$s = c_0\beta = c_0/c'_m = c_0/(2RT)^{1/2} \tag{4.23}$$

is called the *molecular speed ratio.*

For a stationary gas, both s and c_0 are zero and this reduces to

$$\beta\dot{N}_i/n = 1/(2\pi^{1/2})$$

or, from eqn (4.8) and noting that $c = c'$,

$$\dot{N}_i = n\bar{c}/4. \tag{4.24}$$

This result could have been deduced by physical reasoning from the stationary gas version of eqn (4.14). This states that the mean velocity component, taken over the molecules for which it is positive, is equal to half the mean molecular speed $\bar{c}$. Since half the molecules have a positive component in a particular direction, the number flux per unit area per unit volume in a particular direction must be $n\bar{c}/4$.

The *inward normal momentum flux* p_i to the element is obtained by setting $Q = mu = m(u' + c_0\cos\theta)$. i.e.

$$p_i = \frac{nm\beta^3}{\pi^{3/2}} \int_{-\infty}^{\infty} \exp(-\beta^2 w'^2)\,dw' \int_{-\infty}^{\infty} \exp(-\beta^2 v'^2)\,dv' \times \int_{-c_0\cos\theta}^{\infty} (u' + c_0\cos\theta)^2 \exp(-\beta^2 u'^2)\,du'.$$

or

$$\beta^2 p_i/\rho = [s\cos\theta\exp(-s^2\cos^2\theta) + \pi^{1/2}\{1 + \operatorname{erf}(s\cos\theta)\} \times (\tfrac{1}{2} + s^2\cos^2\theta)]/(2\pi^{1/2}). \tag{4.25}$$

For a stationary gas, this gives

$$p_i = \rho/(4\beta^2) = \rho RT/2 = p/2,$$

as would be expected in an equilibrium gas since the inward moving molecules contribute half the pressure. Similarly, the *inward parallel momentum flux* τ_i is obtained for $Q = mv = m(v' + c_0\sin\theta)$. The integral is

$$\tau_i = \frac{nm\beta^3}{\pi^{3/2}} \int_{-\infty}^{\infty} \exp(-\beta^2 w'^2)\,dw' \int_{-\infty}^{\infty} (v' + c_0 \sin\theta) \exp(-\beta^2 v'^2)\,dv'$$
$$\times \int_{-c_0\cos\theta}^{\infty} (u' + c_0\cos\theta) \exp(-\beta^2 u'^2)\,du'$$

and the final result follows as

$$\beta^2 \tau_i/\rho = s \sin\theta\,[\exp(-s^2\cos^2\theta) + \pi^{1/2} s\cos\theta\{1 + \mathrm{erf}(s\cos\theta)\}]/(2\pi^{1/2})$$
$$= s\sin\theta(\beta N_i/n). \tag{4.26}$$

The parallel momentum flux in a stationary equilibrium gas is, of course, zero as a consequence of the symmetry of the distribution function.

Finally, the *inward translational energy flux* $q_{i,tr}$ to the element is obtained by setting $Q = \frac{1}{2}mc^2 = \frac{1}{2}m(u^2 + v^2 + w^2)$ in eqn (4.21) to give

$$\tau_i = \frac{nm\beta^3}{2\pi^{3/2}} \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-c_0\cos\theta}^{\infty} \{(u' + c_0\cos\theta)^2 + (v' + c_0\sin\theta)^2 + w'^2\}$$
$$\times (u' + c_0\cos\theta) \exp\{-\beta^2(u'^2 + v'^2 + w'^2)\}\,du'\,dv'\,dw'$$

or

$$\beta^2 q_{i,tr}/\rho = [(s^2 + 2)\exp(-s^2\cos^2\theta) + \pi^{1/2} s\cos\theta\,(s^2 + 5/2)$$
$$\times\{1 + \mathrm{erf}(s\cos\theta)\}]/(4\pi^{1/2}). \tag{4.27}$$

For a stationary gas, this becomes

$$\beta^3 q_{i,tr}/\rho = 1/(2\pi^{1/2}). \tag{4.28}$$

Unlike the number and momentum flux, the energy flux is modified by the presence of internal energy. For an equilibrium non-reacting gas, the specific internal energy follows from eqn (1.31) as

$$e_{int} = \tfrac{1}{2}\zeta RT.$$

Here, ζ is the number of internal degrees of freedom and may be related to the specific heat ratio of the gas by

$$\gamma = (\zeta + 5)/(\zeta + 3),$$

so that

$$\zeta = (5 - 3\gamma)/(\gamma - 1). \tag{4.29}$$

The inward *internal energy flux* is equal to the product of the number flux N_i and the average internal energy per molecule. Therefore, from eqns (1.31), (4.22), and (4.29),

$$\beta^3 q_{i,int}/\rho = \{(5 - 3\gamma)/(\gamma - 1)\}[\exp(-s^2\cos^2\theta)$$
$$+ \pi^{1/2} s\cos\theta\{1 + \mathrm{erf}(s\cos\theta)\}]/(8\pi^{1/2}). \tag{4.30}$$

The *total energy flux* to the element is obtained by summing eqn (4.27) for $q_{i,tr}$ and eqn (4.30) for $q_{i,int}$, to give

$$\beta^3 q_i/\rho = [\{2s^2 + (\gamma+1)/(\gamma-1)\}[\exp(-s^2\cos^2\theta) + 2\pi^{1/2} s\cos\theta \times \{s^2 + \gamma/(\gamma-1)\}\{1 + \mathrm{erf}(s\cos\theta)\}]/(8\pi^{1/2}). \tag{4.31}$$

The version for a stationary gas is

$$\beta^3 q_i/\rho = \{(\gamma+1)/(\gamma-1)\}/(8\pi^{1/2}). \tag{4.32}$$

The average energy of the molecules crossing the surface element in a stationary monatomic gas is given by q_i/N_i for $s = 0$, or m/β^2. The average energy of a molecule in a spatial element is given by eqn (1.29) as $3mRT/2$ or $\frac{3}{4}m/\beta^2$. The average energy of the molecules crossing a surface element therefore exceeds that in a spatial element by the factor 4/3. The physical explanation for this is that the probability of a fast molecule crossing a surface element in a given time is greater than the corresponding probability for a slower molecule. This is the reason why the total energy flux of eqn (4.32) is exactly twice the translational energy flux of eqn (4.28) when the number of internal degrees of freedom $\zeta = 4$, rather than when $\zeta = 3$.

The above equation for N_i is usually derived in the context of molecular effusion in a free molecule gas. This problem is dealt with in detail in Chapter 5. The other equations for p_i, τ_i, and q_i are generally introduced in the context of free molecule aerodynamics. They then represent the pressure, shear stress, and heat flux incident on an element of solid surface. However, it must be emphasized that the equations apply to the flux quantities across an area element in an equilibrium gas *at any density* and represent important reference quantities.

The general behaviour of the molecular number flux in an equilibrium gas is illustrated in Fig. 4.3 . If $s\cos\theta \to \infty$, then $\exp(-s^2\cos^2\theta) \to 0$ and $\mathrm{erf}(s\cos\theta) \to 1$ so that the hypersonic form of eqn (4.22) is

$$\beta N_i/n = s\cos\theta,$$

or

$$n_i = n c_0 \cos\theta. \tag{4.33}$$

This is the equation that would be derived from first principles if one neglects the thermal velocity components and considers only the flux due to the stream speed. Eqn (4.26) shows that $\beta^2\tau_i/(\rho s\sin\theta)$ is equal to $\beta N_i/n$, so that the behaviour of the parallel momentum flux is also covered by Fig. 4.3. The effect of the thermal velocity components is small at $s\cos\theta = 1$ where $\beta N_i/n = 1.02513$ and negligible at $s\cos\theta = 2$, where $\beta N_i/n = 2.00049$.

The opposite limit of large negative values of $s\cos\theta$ is also important and will be called the 'thermal backflow' limit. The numerical evaluation of eqn (4.22) can be difficult because the net flux is the small difference between two large terms, but this can be overcome by replacing the error function by the series expansion of eqn (B34). The leading term for the thermal backflow limit is

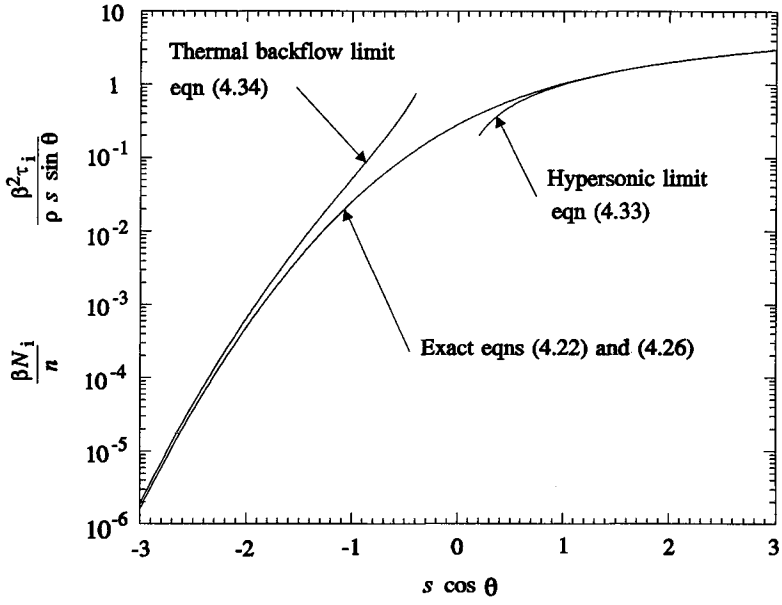


Fig. 4.3 Number and parallel momentum flux to a surface element.

$$\beta N_i/n = \beta^2\tau_i/(\rho s \sin\theta) = \exp(-s^2\cos^2\theta)/(4\pi^{1/2}s^2\cos^2\theta). \qquad (4.34)$$

This limit is also shown in Fig. 4.3 and it can be seen that the thermal backflow limit is approached much more gradually than the hypersonic limit.

Similar plots for the dimensionless normal momentum flux $\beta^2 p_i/\rho$ are presented in Fig. 4.4. Eqn (4.25) shows that the hypersonic limit for this quantity as the exponential and error function terms tend to zero and unity, respectively, is

$$\beta^2 p_i/\rho = \tfrac{1}{2} + s^2\cos^2\theta,$$

or

$$p_i/(\tfrac{1}{2}\rho c_0^2) = 1/s^2 + 2\cos^2\theta. \qquad (4.35)$$

In the context of free-molecule aerodynamics, the incident pressure coefficient tends to a limiting value of $2\cos^2\theta$ as s becomes large in comparison with unity. The departure of the exact result from this is again very small for $s\cos\theta > 2$. The leading term for the incident normal momentum flux in the thermal backflow limit is

$$\beta^2 p_i/\rho = -\exp(s^2\cos^2\theta)/(4\pi^{1/2}s^3\cos^3\theta) \qquad (4.36)$$

and Fig. 4.4 shows that this is again approached more gradually than the hypersonic limit.

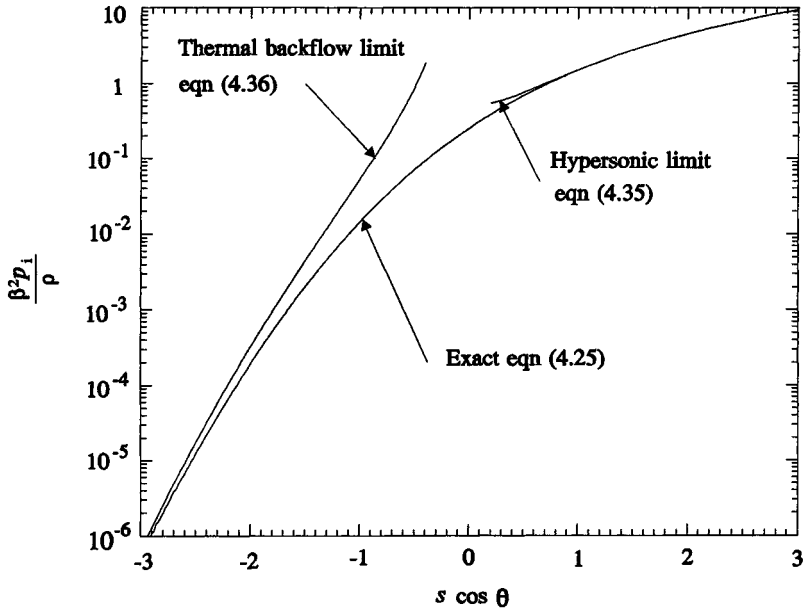


Fig. 4.4 Normal momentum flux to a surface element.

The energy flux is much more complex since it depends on s and θ independently, rather than on $s\cos\theta$, and also depends on the number of internal degrees of freedom. The internal energy flux can be written

$$\beta^3 q_{i,int}/\rho = (\zeta/4)\beta N_i/n \tag{4.37}$$

so that its behaviour is also covered by Fig. 4.3. The hypersonic limit of eqn (4.27) is

$$\beta^3 q_{i,tr}/\rho = \tfrac{1}{2} s\cos\theta(s^2 + 5/2)$$

or

$$q_{i,tr}/(\tfrac{1}{2}\rho c_0^3) = \cos\theta\{1 + 5/(2s^2)\}. \tag{4.38}$$

The behaviour of the translational energy flux as a function of s and θ, separately, is shown in Fig. 4.5. Note that, when s is large compared with unity, the heat transfer depends strongly on the inclination of the element when it is almost aligned with the flow (i.e. when θ is near 90°).

The angular distribution of the molecules crossing a surface element is required for some problems. The integration of eqn (4.4) gives the number of molecules in a stationary gas that have speeds between c and c+dc and an angle between θ and θ+dθ with an axis normal to a surface element as

$$dn = 2\pi^{-\frac{1}{2}} n \beta^3 c^2 \exp(-\beta^2 c^2)\sin\theta d\theta dc.$$

The flux of these molecules from the element is given by the product of this number with the velocity component $c\cos\theta$ normal to the element, i.e.

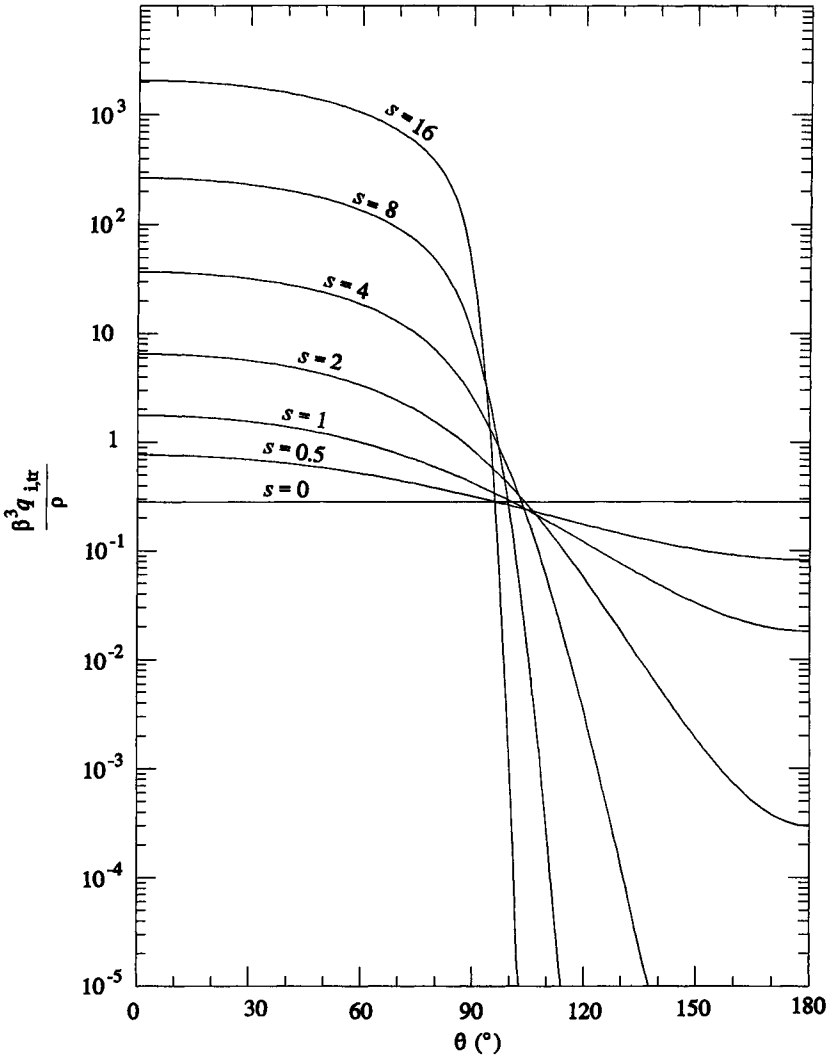


Fig. 4.5 Translational energy flux to a surface element.

$$d\dot{N} = 2\pi^{-\frac{1}{2}} n \beta^3 c^3 \exp(-\beta^2 c^2) \sin\theta \cos\theta \, d\theta \, dc . \tag{4.39}$$

The corresponding two-dimensional result may be obtained from eqn (4.15) by integrating over all values of the axial velocity. The flux of molecules with radial speed between c_n and c_n+dc_n and which make an angle between ϕ and $\phi+d\phi$ with the plane normal to a two-dimensional element is

$$d\dot{N} = \pi^{-1} n \beta^2 c_n^2 \exp(-\beta^2 c_n^2) \cos\phi \, d\phi \, dc_n . \tag{4.40}$$

The equations of this section also apply to each species of a gas mixture.

4.3 Collisional quantities in a simple gas

General expressions were obtained in Chapter 1 for the mean collision rate and mean free path in a dilute gas. These involve the mean value of the product of the total collision cross-section and the magnitude of the relative velocity. The cross-section is generally a function of the relative velocity and the evaluation of these expressions requires the mean value of an arbitrary power j of the relative velocity.

The relative velocity in a binary collision is $\boldsymbol{c}_r = \boldsymbol{c}_1 - \boldsymbol{c}_2$, where the subscripts 1 and 2 denote the two molecules that are involved. Assuming molecular chaos, the two particle distribution function is equal to the product of the two single particle distribution functions f_1 and f_2. The required mean value is then

$$\overline{c_r^j} = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} c_r^j f_1 f_2 \, d\boldsymbol{c}_1 d\boldsymbol{c}_2$$

and, from eqn (4.1) for an equilibrium gas,

$$\overline{c_r^j} = \frac{(m_1 m_2)^{3/2}}{(2\pi k T)^3} \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} c_r^j \exp\{-(m_1 c_1^2 + m_2 c_2^2)/(2kT)\} \, d\boldsymbol{c}_1 d\boldsymbol{c}_2 .$$

This integral is most easily evaluated if the variables are changed from $\boldsymbol{c}_1$ and $\boldsymbol{c}_2$ to $\boldsymbol{c}_r$ and $\boldsymbol{c}_m$. The Jacobian of the transformation is

$$\frac{\partial(u_1, v_1, w_1, u_2, v_2, w_2)}{\partial(u_r, v_r, w_r, u_m, v_m, w_m)}$$

but, because of the symmetry of eqns (2.3) and (2.4), it is sufficient to evaluate the reduced one-dimensional Jacobian. This is, in the x-direction,

$$\frac{\partial(u_1, u_2)}{\partial(u_r, u_m)} = \begin{vmatrix} \dfrac{\partial u_1}{\partial u_r} & \dfrac{\partial u_1}{\partial u_m} \\ \dfrac{\partial u_2}{\partial u_r} & \dfrac{\partial u_2}{\partial u_m} \end{vmatrix}$$

or, using eqn (2.4),

$$\frac{\partial(u_1, u_2)}{\partial(u_r, u_m)} = \begin{vmatrix} \dfrac{m_2}{m_1 + m_2} & 1 \\ -\dfrac{m_1}{m_1 + m_2} & 1 \end{vmatrix} = \frac{m_2}{m_1 + m_2} + \frac{m_1}{m_1 + m_2} = 1 .$$

The full Jacobian is therefore unity and $d\boldsymbol{c}_m \, d\boldsymbol{c}_r$ may be substituted for $d\boldsymbol{c}_1 \, d\boldsymbol{c}_r$ in the above integral. We also have, from eqn (2.6),

$$m_1 c_1^2 + m_2 c_2^2 = (m_1 + m_2) c_m^2 + m_r c_r^2,$$

where m_r is the reduced mass. Since the integrand is independent of the directions of $\boldsymbol{c}_r$ and $\boldsymbol{c}_m$, the velocity space elements may be written in polar coordinates. An integration over all polar and azimuth angles then gives

$$d\boldsymbol{c}_r = 4\pi c_r^2 dc_r \quad \text{and} \quad d\boldsymbol{c}_m = 4\pi c_m^2 dc_m.$$

Therefore,

$$\overline{c_r^j} = \frac{2(m_1 m_2)^{3/2}}{\pi (kT)^3} \int_0^\infty \int_0^\infty c_r^{j+2} c_m^2 \exp[-\{(m_1 + m_2) c_m^2 + m_r c_r^2\}/(2kT)]\, dc_m dc_r,$$

or

$$\overline{c_r^j} = \frac{2(m_1 m_2)^{3/2}}{\pi (kT)^3} \int_0^\infty c_m^2 \exp\{-(m_1 + m_2) c_m^2/(2kT)\}\, dc_m \times \int_0^\infty c_r^{j+2} \exp\{-(m_r c_r^2/(2kT)\}\, dc_r. \tag{4.41}$$

Note that the distribution functions for c_m and c_r can be inferred from eqn (4.41) as

$$f_{c_m} = \frac{4(m_1 + m_2)^{3/2}}{\pi^{1/2} (2kT)^{3/2}} c_m^2 \exp\{-(m_1 + m_2) c_m^2/(2kT)\} \tag{4.42}$$

and

$$f_{c_r} = \frac{4 m_r^{3/2}}{\pi^{1/2} (2kT)^{3/2}} c_r^2 \exp\{-m_r c_r^2/(2kT)\}. \tag{4.43}$$

The application of the standard integrals of Appendix B to eqn (4.43) gives

$$\overline{c_r^j} = (2/\pi^{1/2})\, \Gamma\{(j+3)/2\} (2kT/m_r)^{j/2}. \tag{4.44}$$

The mean value of the relative speed is obtained by setting j=1, i.e.

$$\overline{c_r} = (2/\pi^{1/2})(2kT/m_r)^{1/2}. \tag{4.45}$$

For a simple gas, $m_r = m/2$, and eqn (4.8) may be used to relate this value to the mean thermal speed of an individual molecule. The result is

$$\overline{c_r} = 2^{3/2}/(\pi^{1/2}\beta) = 2^{1/2}\overline{c'}, \tag{4.46}$$

Note that this is also the mean value of the relative speed *in collisions* only for the special case of hard sphere molecules. The total cross-section of the other molecular models is a function of the relative speed and the mean value in collisions is affected by this function. The mean collision rate and mean free path in an equilibrium gas also depend on the molecular model.

Eqn (4.44) may be applied to the combination of eqn (1.10) for the collision rate and the definition in eqns (3.60) and (3.67) of the VHS model to give the mean collision rate per molecule in an equilibrium gas of VHS or VSS molecules as

$$\nu_0 = n\sigma_{T,ref} c_{r,ref}^{2\omega-1} \frac{2}{\pi^{1/2}} \Gamma(5/2-\omega)\left(\frac{2kT}{m_r}\right)^{1-\omega}. \tag{4.47}$$

The preceding analysis does not apply to the GHS molecular model because its cross-section is not a power law of the relative speed. However, the mean value of the product of their relative speed and the cross-section can be written

$$\overline{\sigma_T c_r} = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \sigma_T c_r f_1 f_2 \, d\boldsymbol{c}_1 d\boldsymbol{c}_2$$

$$= \frac{(m_1 m_2)^{3/2}}{(2\pi kT)^3} \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \sigma_T c_r \exp\{-(m_1 c_1^2 + m_2 c_2^2)/(2kT)\} d\boldsymbol{c}_1 d\boldsymbol{c}_2 .$$

The variables may be transformed in exactly the same way as for the VHS and VSS molecules and the result can be written as

$$\overline{\sigma_T c_r} = (2/\pi)^{1/2}\{m_r/(kT)\}\int_0^{\infty} \sigma_T c_r^3 \exp\{-m_r c_r^2/(2kT)\} \, dc_r . \tag{4.48}$$

The GHS cross-section is given by eqn (3.80) and eqn (4.48) may then be integrated to give the collision rate in an equilibrium GHS gas as

$$\nu_0 = 2s_0^2\left(\frac{2kT}{\pi m_r}\right)^{1/2}\left(\alpha_1(kT/\varepsilon)^{-\psi_1}\Gamma(2-\psi_1) + \alpha_2(kT/\varepsilon)^{-\psi_2}\Gamma(2-\psi_2)\right). \tag{4.49}$$

Eqn (3.73) enables the reference values in eqn (4.47) for the VSS molecules to be written in terms of the coefficient of viscosity. We then have

$$\nu_0 = \frac{5(\alpha+1)(\alpha+2)}{\alpha(5-2\omega)(7-2\omega)} \frac{p}{\mu}. \tag{4.50}$$

The number of collisions per unit time per unit volume follows from eqn (1.11) as

$$N_{c,0} = \frac{5(\alpha+1)(\alpha+2)}{2\alpha(5-2\omega)(7-2\omega)} \frac{np}{\mu}. \tag{4.51}$$

Eqns (1.12) and (4.8) then give the equilibrium mean free path as

$$\lambda_0 = \frac{4\alpha(5-2\omega)(7-2\omega)}{5(\alpha+1)(\alpha+2)}\left(\frac{m}{2\pi kT}\right)^{1/2}\frac{\mu}{\rho} = \frac{4\alpha(5-2\omega)(7-2\omega)}{5\pi^{1/2}(\alpha+1)(\alpha+2)}\frac{\beta\mu}{\rho}. \tag{4.52}$$

The simple hard sphere results are obtained if $\omega = \frac{1}{2}$ and $\alpha = 1$. Because the total cross-section is constant, the equilibrium collision rate can be written directly from eqns (4.45) and (1.10) as

$$\nu_0 = 2^{\frac{1}{2}}\sigma_T n\overline{c'} = (4/\pi^{\frac{1}{2}})\sigma_T n(kT/m)^{\frac{1}{2}}. \tag{4.53}$$

The number of collisions per unit time per unit volume again follows from eqn (1.11) as

$$N_{c,0} = 2^{-\frac{1}{2}}\sigma_T n^2\overline{c'}, \tag{4.54}$$

and the equilibrium mean free path follows from eqns (4.53) and (1.12) as

$$\lambda_0 = (2^{\frac{1}{2}}\sigma_T n)^{-1} = (2^{\frac{1}{2}}\pi d^2 n)^{-1}. \tag{4.55}$$

This can also be written from eqn (4.52) as

$$\lambda_0 = \frac{16}{5}\left(\frac{m}{2\pi kT}\right)^{\frac{1}{2}}\frac{\mu}{\rho}. \tag{4.56}$$

The hard sphere equations have been widely used to derive an effective diameter for real molecules and to define the mean free path in both numerical and analytical studies. These procedures are generally carried out at a single temperature and, if the results are then applied over a range of temperatures, errors will be introduced. Consistent correlations and accurate calculations require a molecular model that has a cross-section that changes with the relative collision velocity in a similar manner to the changes that occur in the real gas. As noted earlier, this is achieved if the model has a temperature dependence of the coefficient of viscosity that matches that in the real gas. The traditional hard sphere equations must therefore be replaced by those based on a more realistic model, and the VSS model is the obvious choice. This also permits the definition of a more realistic temperature-dependent diameter and Knudsen number.

The VSS result for the mean free path in eqn (4.52) differs from the conventional eqn (4.56) that employs the hard sphere model only by the numerical factor $\alpha(5-2\omega)(7-2\omega)/\{4(\alpha+1)(\alpha+2)\}$. This factor is, of course, unity for hard sphere molecules which have $\omega = \frac{1}{2}$ and $\alpha = 1$, but decreases to 0.8244 for the more realistic $\omega = \frac{3}{4}$ and $\alpha = 1.5325$, and to 0.6173 for $\omega = 1$ and $\alpha = 2.1403$ which corresponds to the limiting case of Maxwell molecules.

This relationship between the mean free path and the coefficient of viscosity enables the Chapman–Enskog distribution function to be written in terms of the local Knudsen numbers. The substitution of eqn (4.52) into eqn (3.87), and using eqn (3.56), gives

$$f = f_0\left[1 - \frac{5\pi^{\frac{1}{2}}(\alpha+1)(\alpha+2)\beta v'}{4\alpha(5-2\omega)(7-2\omega)} \times\left(3(\beta^2 c'^2 - 5/2)\frac{\lambda}{T}\frac{\partial T}{\partial y} + 4\beta u's\frac{\lambda}{u_0}\frac{\partial u_0}{\partial y}\right)\right]. \tag{4.57}$$

The implications of this equation for the range of validity of the Navier–Stokes equations were discussed in Chapter 1.

The mean value, over all collisions in a VHS or VSS gas, of any quantity Q that is a function of c_r alone, may be obtained by setting j to the value appropriate to the product of total cross-section and relative speed, and then including Q within the integral over c_r in eqn (4.41) and then dividing the resulting expression by eqn (4.44), i.e.

$$\bar{Q} = \{2/\Gamma(5/2-\omega)\}\{m_r/(2kT)\}^{5/2-\omega} \times \int_0^\infty Q c_r^{2(2-\omega)} \exp\{-m_r c_r^2/(2kT)\} dc_r. \tag{4.58}$$

In the case of the GHS gas, the mean value over all collisions of any quantity Q is given by

$$\bar{Q} = \frac{\int_0^\infty Q \sigma_T c_r^3 \exp\{-m_r c_r^2/(2kT)\} dc_r}{\int_0^\infty \sigma_T c_r^3 \exp\{-m_r c_r^2/(2kT)\} dc_r}$$

or, using eqn (4.48)

$$\bar{Q} = (2/\pi)^{1/2}\{m_r/(kT)\}^{3/2} \int_0^\infty Q \sigma_T c_r^3 \exp\{-m_r c_r^2/(2kT)\} dc_r / \overline{\sigma_T c_r}. \tag{4.59}$$

The collision cross-section in a VSS gas changes with the magnitude of the relative velocity in the collision, and eqns (3.60) and (3.74) show that it is inversely proportional to c_r to the power $2\omega-1$. A temperature-based effective or reference diameter d_{ref} is most conveniently defined as the diameter in a VSS gas in equilibrium at the mean value of $c_r^{2\omega-1}$. This diameter is related to the earlier reference cross-section of eqn (3.60) by

$$\sigma_{ref} = \pi d_{ref}^2 = \sigma_{T,ref} c_{r,ref}^{2\omega-1} / \overline{c_r^{2\omega-1}}. \tag{4.60}$$

Eqn (4.58) may be evaluated with $Q = c_r^{2\omega-1}$ to give

$$\overline{c_r^{2\omega-1}} = (2kT/m_r)^{\omega-1/2} / \Gamma(5/2-\omega). \tag{4.61}$$

Eqns (4.60) and (4.61) may then be combined with eqns (3.67) and (3.73) to give the reference diameter of a gas that has, at temperature T_{ref}, a coefficient of viscosity μ_{ref} with temperature exponent ω, i.e.

$$d_{ref} = \left(\frac{5(\alpha+1)(\alpha+2)(mkT_{ref}/\pi)^{1/2}}{4\alpha(5-2\omega)(7-2\omega)\mu_{ref}} \right)^{1/2}. \tag{4.62}$$

The average cross-section based on this relation is larger than the hard sphere cross-section that would be deduced from eqn (3.59) by the factor $4(\alpha+1)(\alpha+2)/\{\alpha(5-2\omega)(7-2\omega)\}$. This result is consistent with the earlier

finding for the mean free path and means that the hard sphere result of eqn (4.56) applies also to the VHS gas, as long as the cross-section is based on this reference diameter.

The VSS model may be implemented either through eqn (3.74), which makes no use of equilibrium gas results, or through this new reference diameter. The latter option has the advantage of avoiding the appearance of the viscosity coefficient in relations as a reference quantity, even when viscosity is physically irrelevant.

Eqn (4.60) can then be written

$$\pi d_{\mathrm{ref}}^2 \overline{c_{\mathrm{r}}^{2\omega-1}} = \sigma_{\mathrm{T,ref}} c_{\mathrm{r,ref}}^{2\omega-1} = \pi d^2 c_{\mathrm{r}}^{2\omega-1},$$

where d is the diameter at relative speed c_{r}. Therefore, using eqn (4.61),

$$d = d_{\mathrm{ref}} [\{2kT_{\mathrm{ref}}/(m_{\mathrm{r}} c_{\mathrm{r}}^2)\}^{\omega-\frac{1}{2}} / \Gamma(5/2-\omega)]^{\frac{1}{2}}. \tag{4.63}$$

Some typical values of the reference diameter from eqn (4.62) at a reference temperature of 0°C, are presented in Appendix A. The equation that corresponds to eqn (4.63) and the reference diameters that appear in the original presentation of the VHS model (Bird 1981) differ from those defined here by a small numerical factor. This is because they were based on the mean value of c_{r}^2 raised to the power $\omega-\frac{1}{2}$, rather than the mean value of $c_{\mathrm{r}}^{2\omega-1}$. The degree of arbitrary choice in the definition of a reference diameter is of no consequence because it can be eliminated from eqns (4.62) and (4.63). This recovers eqn (3.74) and shows that the model is independent of d_{ref} and does not require an equilibrium gas.

The introduction of the reference diameter and the combination of eqn (4.63) with eqn (1.10) and (4.44) enables the equilibrium collision rate per molecule in a VHS or VSS gas to be written

$$\nu_0 = 4 d_{\mathrm{ref}}^2 n (\pi k T_{\mathrm{ref}}/m)^{\frac{1}{2}} (T/T_{\mathrm{ref}})^{1-\omega}. \tag{4.64}$$

Similarly, the equilibrium mean free path may be written

$$\lambda_0 = \{2^{\frac{1}{2}} \pi d_{\mathrm{ref}}^2 n (T_{\mathrm{ref}}/T)^{\omega-\frac{1}{2}}\}^{-1}. \tag{4.65}$$

Note that the collision rate is independent of temperature in a Maxwell gas ($\omega = 1$), while the mean free path is independent of temperature in a hard sphere gas ($\omega = \frac{1}{2}$). The value of ω in real gases is between these limits, and both the collision rate and mean free path depend on the temperature. The translational kinetic energy in the centre of mass frame of reference is $E_{\mathrm{t}} = \frac{1}{2} m_{\mathrm{r}} c_{\mathrm{r}}^2$, and the substitution of $Q = \frac{1}{2} m_{\mathrm{r}} c_{\mathrm{r}}^2$ in eqn (4.59) gives

$$\overline{E_{\mathrm{t}}} = (5/2-\omega) kT \tag{4.66}$$

for the mean translational kinetic energy in collisions. For the special case of hard sphere molecules, this reduces to

$$\overline{E_{\mathrm{t}}} = 2kT. \tag{4.67}$$

The fraction of collisions in which $E_t = \frac{1}{2}m_r c_r^2$ exceeds some reference value E_m is important in the context of chemical reactions. The total number of collisions is proportional to the integral in eqn (4.43) and the number of these with $E_t > E_m$ is obtained by setting the lower limit for the integration over c_r to $(2E_m/m_r)^{\frac{1}{2}}$. The fraction then becomes

$$\frac{dN}{N} = \frac{2}{\Gamma(5/2-\omega)}\left(\frac{m_r}{2kT}\right)^{5/2-\omega}\int_{(2E_m/m_r)^{\frac{1}{2}}}^{\infty} c_r^{2(2-\omega)}\exp\left(-\frac{m_r c_r^2}{2kT}\right)dc_r \qquad (4.68)$$

or, if the incomplete gamma function is introduced,

$$dN/N = \Gamma\{5/2-\omega, E_m/(kT)\}/\Gamma(5/2-\omega)\,. \qquad (4.69)$$

For hard sphere molecules, $\omega=\frac{1}{2}$ and the reduction formula of eqn (B6) enables this to be written

$$dN/N = \exp\{-E_m/(kT)\}\{E_m/(kT)+1\}\,. \qquad (4.70)$$

This fraction is higher than that for the more realistic molecules which have collision cross-sections that decrease as the relative translational energy increases. The fraction is shown in Fig. 4.6 as a function of the relative translational energy for the hard sphere model ($\omega=0.5$), the more realistic case with $\omega=0.75$, and the limiting 'soft molecule' case with $\omega=1$.

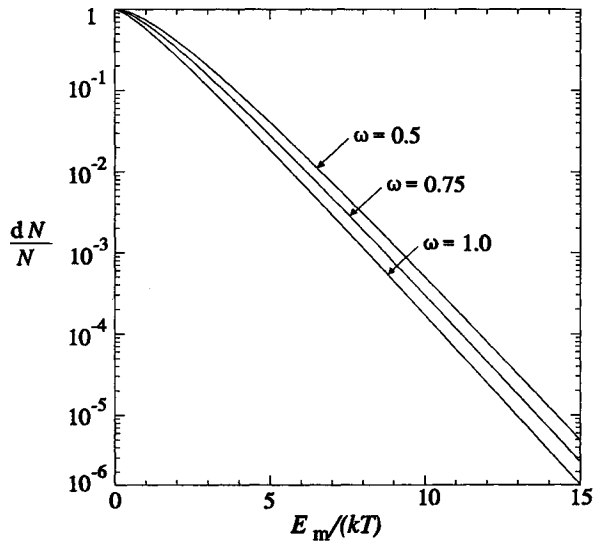


Fig. 4.6 The fraction of collisions with relative translational energy in excess of $E_m/(kT)$.

An important variation of this result occurs when one is interested only in the energy based on the component of the relative velocity along the apse line of the collision or, for hard sphere or VHS molecules, along the line of centres. This means that, for a given c_r, the angle θ_A must be between 0 and $\cos^{-1}\{(2E_m/m_r)^{1/2}/c_r\}$. The differential cross-section for VHS molecules is

$$\sigma d\Omega = \{\sigma_T/(4\pi)\} \sin\chi \, d\chi \, d\varepsilon,$$

and eqn (2.21) shows that

$$\sin\chi \, d\chi = -4 \sin\theta_A \cos\theta_A \, d\theta_A .$$

The effective total collision cross-section σ_E of the collisions with the required relative translational energy is, therefore,

$$\sigma_E = -2\sigma_T \int_0^{\cos^{-1}\{(2E_m/m_r)^{1/2}/c_r\}} \sin\theta_A \cos\theta_A \, d\theta_A ,$$

or, since $\sin\theta_A \, d\theta_A = -d(\cos\theta_A)$,

$$\sigma_E = \sigma_T\left(1 - \frac{2E_m}{m_r c_r^2}\right). \tag{4.71}$$

The required fraction is therefore obtained by incorporating the factor in parentheses in eqn (4.71) into eqn (4.68), i.e.

$$\frac{dN}{N} = \frac{2}{\Gamma(5/2-\omega)}\left(\frac{m_r}{2kT}\right)^{5/2-\omega} \times \int_{(2E_m/m_r)^{1/2}}^{\infty} c_r^{2(2-\omega)}\left(1 - \frac{2E_m}{m_r c_r^2}\right) \exp\left(-\frac{m_r c_r^2}{2kT}\right) dc_r ,$$

which may be evaluated to give

$$\frac{dN}{N} = \frac{1}{\Gamma(5/2-\omega)}\left[\Gamma\left(\frac{5}{2}-\omega, \frac{E_m}{kT}\right) - \frac{E_m}{kT}\Gamma\left(\frac{3}{2}-\omega, \frac{E_m}{kT}\right)\right]. \tag{4.72}$$

For the special case of a hard sphere gas, this reduces to give the strikingly simple result

$$dN/N = \exp\{-E_m/(kT)\} \tag{4.73}$$

for the fraction of collisions in which the relative translational energy along the line of centres exceeds the value E_m.

4.4 Collisional quantities in a gas mixture

Eqns (1.36)–(1.41) are the general results for the collision rates and the mean free path in a gas mixture. The preliminary equations in the preceding section allowed for collisions between molecules of different mass, and it was pointed out in §3.5 that it is desirable to define separate values of the collisional parameters for each combination of species. For a gas of VHS or VSS molecules, these parameters are the reference diameter, the reference temperature, and the temperature exponent ω.

Consider a mixture of s separate gas species of VHS molecules for which an s by s array of reference quantities is available. The analysis that led to the simple gas result of eqn (4.2) is readily modified to give the collision rate of a species p molecule with molecules of species q in an equilibrium gas. The reduced mass for this combination is used together with the appropriate reference parameters to give

$$(\nu_{pq})_0 = 2\pi^{\frac{1}{2}}(d_{\text{ref}})_{pq}^2 n_q \{T/(T_{\text{ref}})_{pq}\}^{1-\omega_{pq}} \{2k(T_{\text{ref}})_{pq}/m_r\}^{\frac{1}{2}}. \tag{4.74}$$

The mean collision rate for species p is obtained from eqn (1.37) as

$$(\nu_p)_0 = \sum_{q=1}^{s} \left(2\pi^{\frac{1}{2}}(d_{\text{ref}})_{pq}^2 n_q \{T/(T_{\text{ref}})_{pq}\}^{1-\omega_{pq}} \{2k(T_{\text{ref}})_{pq}/m_r\}^{\frac{1}{2}} \right). \tag{4.75}$$

The mean collision rate for the mixture follows from eqn (1.38), and the number of collisions per unit time per unit volume from eqn (1.11).

The mean free path of a species p molecule is equal to its mean thermal velocity divided by its collision rate. This quantity is obtained from eqns (4.8) and (4.75) as,

$$(\lambda_p)_0 = \left[\sum_{q=1}^{s} \left\{ \pi (d_{\text{ref}})_{pq}^2 n_q \left(\frac{(T_{\text{ref}})_{pq}}{T} \right)^{\omega_{pq}-\frac{1}{2}} \left(1 + \frac{m_p}{m_q} \right)^{\frac{1}{2}} \right\} \right]^{-1}. \tag{4.76}$$

Finally, the overall mean free path for this mixture is

$$\lambda_0 = \sum_{p=1}^{s} \frac{n_p}{n} \left[\sum_{q=1}^{s} \left\{ \pi (d_{\text{ref}})_{pq}^2 n_q \left(\frac{(T_{\text{ref}})_{pq}}{T} \right)^{\omega_{pq}-\frac{1}{2}} \left(1 + \frac{m_p}{m_q} \right)^{\frac{1}{2}} \right\} \right]^{-1}. \tag{4.77}$$

It is, therefore, not an easy matter to evaluate the exact Knudsen number for problems that involve the flow of gas mixtures.

A result that is required for future reference is the number of collisions per unit time per unit volume in an equilibrium gas between molecules of species p and molecules of species q. This result follows from eqn (4.74) as

$$(N_{pq})_0 = 2\pi^{\frac{1}{2}}(d_{\text{ref}})_{pq}^2 n_p n_q \{T/(T_{\text{ref}})_{pq}\}^{1-\omega_{pq}} \{2k(T_{\text{ref}})_{pq}/m_r\}^{\frac{1}{2}}. \tag{4.78}$$

Note that this expression counts all collisions twice over in the sense that a single collision event advances both N_{pq} and N_{qp}. A symmetry factor of ½ must be applied to recover the result for the number of collision events in a simple gas.

The effective diameter that determines the collision cross-section at relative speed c_r follows from eqn (4.63) as

$$d_{pq} = (d_{\text{ref}})_{pq} \left[\{ 2k(T_{\text{ref}})_{pq} / (m_r c_r^2) \}^{\omega - \frac{1}{2}} / \Gamma(5/2 - \omega_{pq}) \right]^{\frac{1}{2}}. \tag{4.79}$$

The average relative translational energy in collisions between VHS molecules follows from a similar extension of the simple gas theory. i.e.

$$\overline{(E_{\text{tr}})_{pq}} = (5/2 - \omega_{pq}) k (T_{\text{ref}})_{pq}. \tag{4.80}$$

To extend this result to the whole gas again requires a summation and averaging process that also involves the collision rate for each combination of molecules.

4.5 Equilibrium with a solid surface

Maxwell (1879) proposed two models for the interaction of a stationary equilibrium gas with a solid surface that maintain equilibrium. *Specular reflection* is perfectly elastic with the molecular velocity component normal to the surface being reversed, while those parallel to the surface remain unchanged. In *diffuse reflection*, the velocity of each molecule after reflection is independent of its initial velocity. However, the velocities of the reflected molecules as a whole are distributed in accordance with the half-range equilibrium or Maxwellian distribution for the molecules that are directed away from the surface. Equilibrium diffuse reflection requires that both the surface temperature and the temperature associated with the reflected gas Maxwellian be equal to the gas temperature. In the case of specular reflection, the gas may have a stream velocity parallel to the surface, and a specularly reflecting surface is functionally identical to a plane of symmetry.

The general requirement for equilibrium between a gas and solid surface at the molecular level is that the interaction should satisfy the *reciprocity condition*. This is a relationship between the probability of a gas-surface interaction with a particular set of incident and reflected velocities and the probability of the inverse interaction. It may be written (Cercignani 1969, Wenaas 1971, or Kuščer 1971) as

$$\begin{aligned} \boldsymbol{c}_r . \boldsymbol{e} P(-\boldsymbol{c}_r, -\boldsymbol{c}_i) \exp\{-E_{c_r}/(kT_w)\} \\ = -\boldsymbol{c}_i . \boldsymbol{e} P(\boldsymbol{c}_i, \boldsymbol{c}_r) \exp\{-E_{c_i}/(kT_w)\}. \end{aligned} \tag{4.81}$$

The unit vector $\boldsymbol{e}$ has been taken normal to the surface, which is at temperature T_w. $P(\boldsymbol{c}_1, \boldsymbol{c}_2)$ is the probability that a molecule incident on the surface with velocity $\boldsymbol{c}_1$ leaves with velocity $\boldsymbol{c}_2$, and E_c is the energy of the molecule. This condition is related to the principle of detailed balance, and is satisfied by both the diffuse and specular models for a gas in equilibrium with a surface.

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